multiple req at a time. ← **RAID** → Redundant Arrays of I (independent → theory / Inexpensive → practical) disk

Our goal

magnetic disk : opposite SLED

Single Large expensive Disk.

small part where data should be distributed → 1 req at a time

Why redundant? → if host is corrupted

crucial / critical data. → wise to make vulnerable data multiple copies. Continuously available, speedily recoverable.

OS will decide which data should be redundent.

Physically Redundant.



represent / appear as single

AMS → Array management software.

OS → store / process or retrieve



n → 4 Strip / Stripe

physical existance

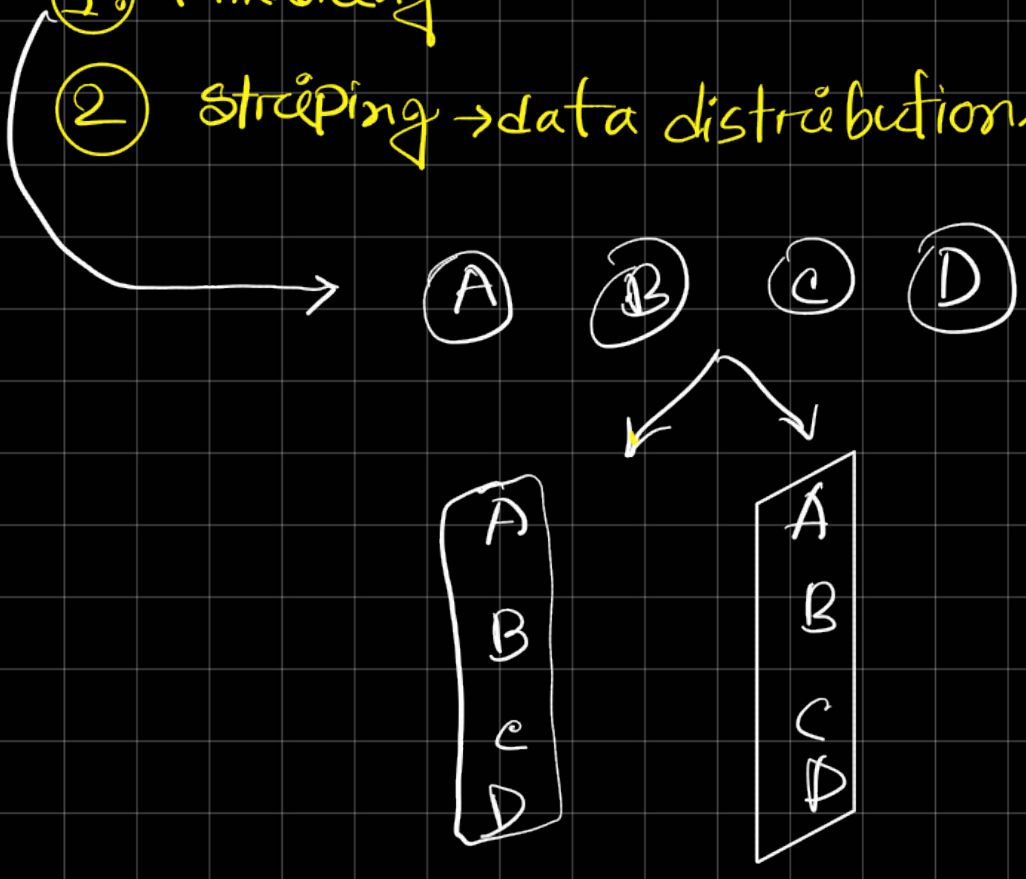
of unique block of data

A single where all are evenly distributed.

To Apply RAID operation

2 method:

1. Mirroring
2. Striping → data distribution.

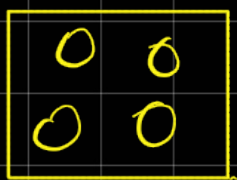


RAID

7 levels. No hierarchical order problem in prev level solved in next.

0-6.

2. logically



OS will see only 1.

(1) Physically

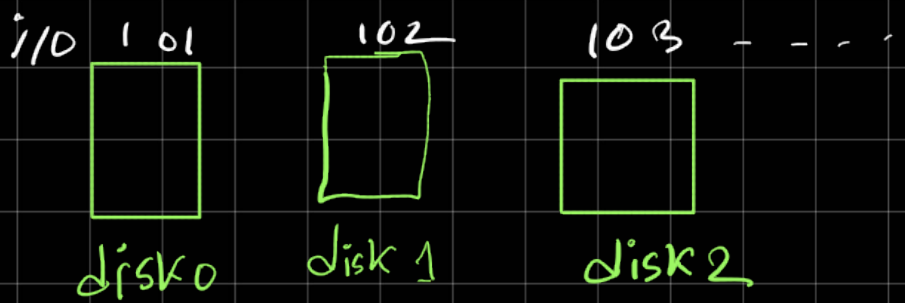
(3) Speedily recoverable.

Not index! particular type → distribution / Striping.

RAID 0 → no / very basic feature of RAID

As it doesn't have all → partial RAID

No Redundancy



independently serve i/o req.

Must → Multiple channel must fit in / simultaneously.

so Increase width / bandwidth.

RAID 1 → Redundancy 1.

logically 4 disk → so for R0 → 4 disk

" R1 → 8 disk (mirror so double) ✓

mirror strip by strip. problem → doubled space.

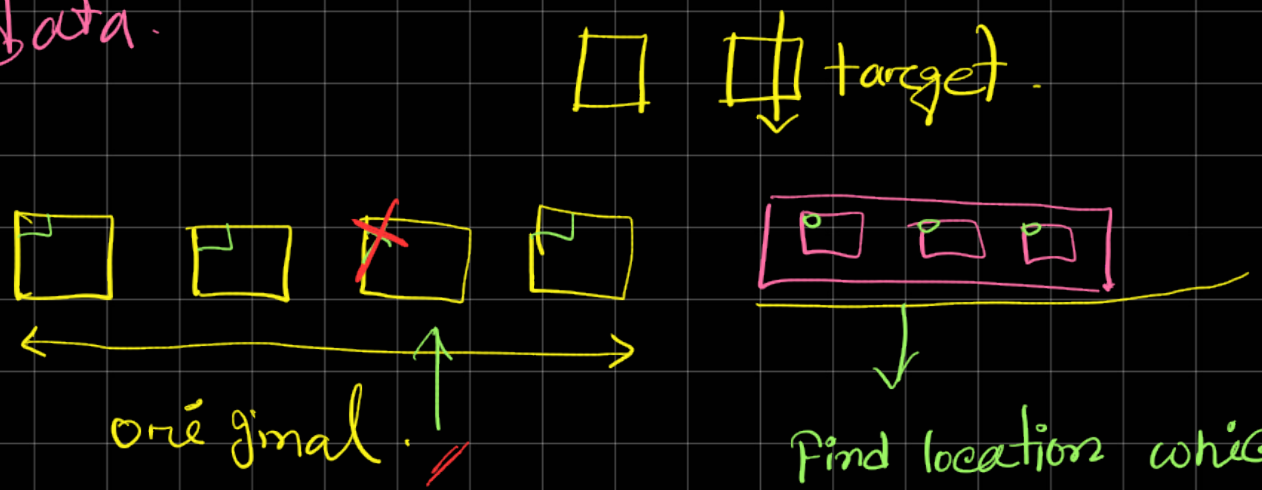
logically 16 → for R0 → 4 disk

"RAID → 8"

RAID 2 reduce redundancy:

Hamming / Ecc.

0~15 → same no. of strips as original data.



$m = 4$

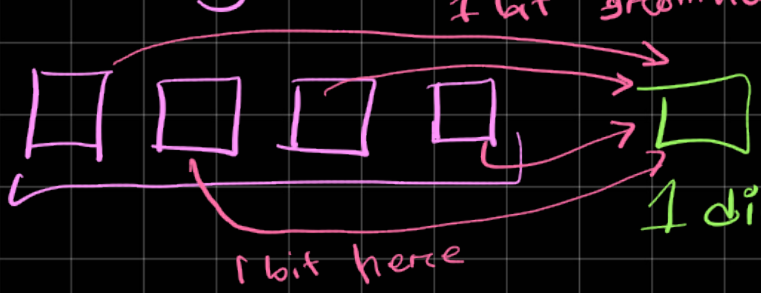
$k = ?3$

$m + k < 2^k + 1$

$4 - 1$

generate bit by bit. bit by bit stored

RAID 3 Redundancy Reduced.



4 bit parity

1 disk for recovering any of the 4 disk

You know which disk is corrupted

if u don't, go to RAID 2.

Parity bit will give signal whether the disk is corrupted or not. (verification)

problem for both R2 & R3 → Write penalty.

Reading done successfully. how to write? Which disk to update → ALL that are involved in generating

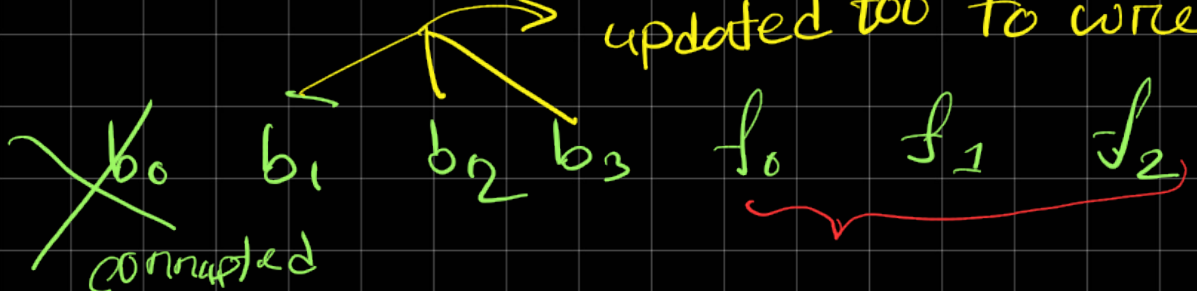
total R & W OP performed

Write penalty:

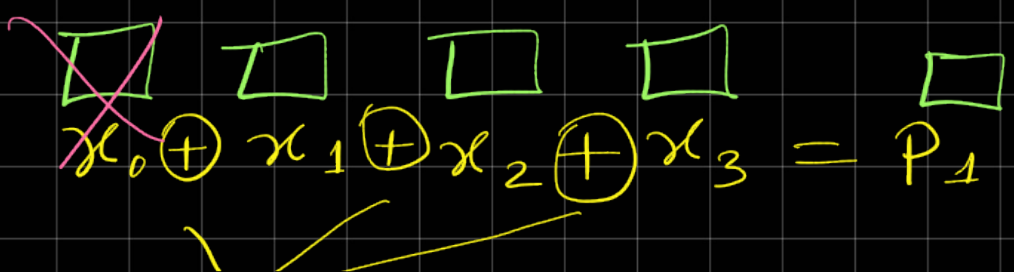
R1 → 2

R2 → at least 7 → updating f_0, f_1, f_2

→ updated too to write new.



R3 → 2



XOR

$x_0' \oplus x_1 \oplus x_2 \oplus x_3 = P_1'$

1100
1100
0000

$x_0 \oplus x_0 \oplus x_0' \oplus x_1 \oplus x_2 \oplus x_3 = P_1'$

same.

0

$0 \oplus 0 \to 0$

$1 \oplus 0 \to 1$

P_1

$x_0 \oplus x_0 \oplus x_0 \oplus x_1 \oplus x_2 \oplus x_3 = P_1'$

$x_0 \oplus x_0' \oplus P_1 = P_1$

2 disks need to be updated.

Strips smaller.

Disadvantage:

① all disk will be involved to perform a single I/O req

Adv

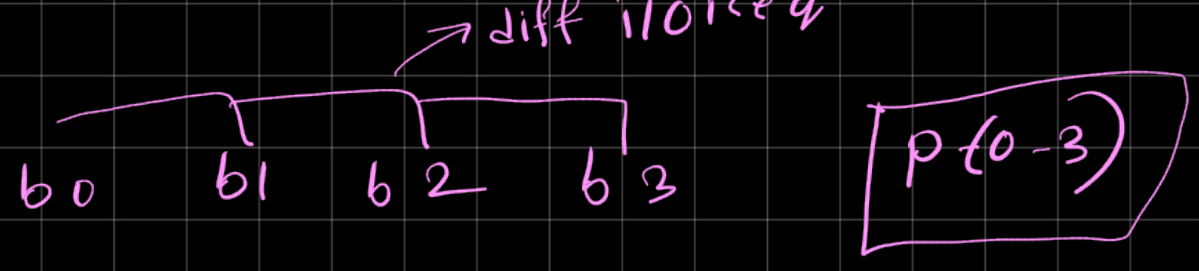
① bandwidth ↑

i.e. from P4 → P6 ∘ Larger strips

to store large amount of data.

↳ so block not strip.

RAID 4 : Multiple write operation:



for write op. all block will knock single Parity block. can't write multiple ones simultaneously.

RAID 5 : Distributed parity blocks

b2 ask → disk 5

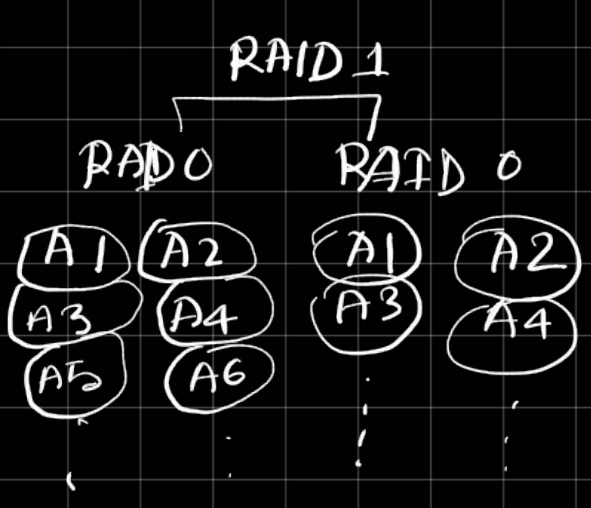
b5 ll → disk 4

RAID 6 Detect 2 different disk corruption.

galloisis field coefficient.

Nested RAID LEVEL

RAID 0+1



mirror o Stripe

RAID 1+0

RAID 0

RAID 1 RAID 1

A1 A1 A2 A2

A3 A3 A4 A4

Stripe of Mirrors

SSD → Solid state Drive